SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Transformer Mechanics Worksheet

Course Code:	CSA6502	**Course Title:**	Generative AI and Large Scale Model
CO Assessed:	CO1-Imitation (BL3)	**Assessment:**	Assessment Tool 3– **Transformer Mechanics Worksheet**
Weightage:	10% of CIA	**Total Marks:**	20
CO1 Statement:	Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models.		
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Section / Batch:	CSE AI	**Date:**	21/07/2026

Code of Conduct

I N.Somashekar Naidu ***/ 192472347*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: ____ N.Somashekar Naidu __________________

Instructions

- This assessment contains 4 set of questions.
- Each student assigned with one set of questions .
- Each question carries 5 marks. Total: 20 Marks.
- Ensure clarity, structure, and logical flow in diagrams.
- Duration: 20 Minutes.

Annexure A Transformer Mechanics Worksheet

Question 1: Tokenization (5 marks). The word "UNHAPPINESS" is uncommon enough that a sub-word tokenizer would likely split it into smaller pieces rather than keep it whole.

(a) Split "UNHAPPINESS" into 3 reasonable sub-word pieces (there's no single "correct" answer — justify your split).

(b) In 1-2 sentences, explain WHY sub-word tokenization (rather than whole-word tokenization) helps a model handle a word like this one, even if the model has never seen the whole word before in training.

Question 2: Embeddings (5 marks). Consider the word pairs “cat” and “kitten” and “cat” and “car”.

(a) Which pair would you expect to have HIGHER cosine similarity in embedding space, and why?

(b) In your own words, what does an embedding actually represent, and how is it different from a one-hot vector?

Question 3: Self-Attention Worked Example (5 marks). Read the sentence: "The trophy didn't fit in the suitcase because it was too big."

(a) The word "it" should attend most strongly to which other word in the sentence? Circle one: "trophy" / "suitcase"

(b) Explain in 2-3 sentences WHY self-attention (not a simple word-order rule) is needed to correctly resolve what "it" refers to here.

Question 1: Tokenization

(a) Split "UNHAPPINESS" into 3 reasonable sub-word pieces. Justify your split.

Word: UNHAPPINESS

Sub-word Split:

1. UN
2. HAPPY
3. NESS

Justification:

- **UN** is a **prefix** that means **"not"** or **"opposite of."** It changes the meaning of the root word into its opposite.
- **HAPPY** is the **root word** that expresses the feeling of joy or pleasure.
- **NESS** is a **suffix** used to convert an adjective into a noun, representing a state or condition.

Therefore,

UN + HAPPY + NESS = UNHAPPINESS

This combination means **"the state of not being happy."**

(b) Explain why sub-word tokenization (rather than whole-word tokenization) helps a model handle a word like this one, even if the model has never seen the whole word before in training.

Sub-word tokenization breaks a word into smaller meaningful units such as prefixes, root words, and suffixes instead of treating the entire word as a single token. Since the model has likely learned the meanings of **"UN,"** **"HAPPY,"** and **"NESS"** separately during training, it can combine these pieces to understand the meaning of **"UNHAPPINESS"** even if the complete word was never present in the training data.

This approach also reduces the number of unknown words (Out-of-Vocabulary words) and helps the model understand rare, new, or complex words more effectively.

Question 2: Embeddings

(a) Which pair would you expect to have HIGHER cosine similarity in embedding space, and why?

Given Word Pairs:

- **cat** and **kitten**
- **cat** and **car**

The pair **"cat" and "kitten"** would have the **higher cosine similarity**.

Reason:

Word embeddings place words with similar meanings or contexts close together in a high-dimensional vector space.

- A **kitten** is a young **cat**.
- Both words belong to the same category of animals.
- They frequently appear in similar contexts such as pets, animals, food, playing, and veterinary care.

Therefore, their embedding vectors point in nearly the same direction, producing a **high cosine similarity value**.

On the other hand,

- A **car** is a vehicle.
- It has no semantic relationship with a **cat**.
- Their contexts are completely different.

Hence, **cat** and **car** have a much lower cosine similarity.

(b) In your own words, what does an embedding actually represent, and how is it different from a one-hot vector?

An **embedding** is a dense numerical vector that represents the meaning of a word based on how it is used in language. It captures semantic relationships between words, so words with similar meanings have similar vectors.

For example,

- **cat**
- **dog**
- **kitten** will have embeddings that are close to one another because they are semantically related.

A **one-hot vector**, however, is simply a unique identifier for each word. It contains one position with the value **1** and all other positions as **0**. It does not contain any information about the word's meaning or relationship to other words.

Difference between Embedding and One-Hot Vector

Embedding	**One-Hot Vector**
Dense numerical vector	Sparse binary vector
Learns semantic meaning	Only identifies the word
Similar words have similar vectors	Every word is equally different
Lower dimensional	Very high dimensional
Used in modern NLP models	Used mainly for basic word representation

Question 3: Self-Attention Worked Example

Sentence:

"The trophy didn't fit in the suitcase because it was too big."

(a) The word "it" should attend most strongly to which other word in the sentence?

trophy

(b) Explain in 2–3 sentences why self-attention (not a simple word-order rule) is needed to correctly resolve what "it" refers to here.

Self-attention examines the relationships between all the words in the sentence instead of relying only on their positions. In this sentence, the phrase **"too big"** explains why the object could not fit inside the suitcase, so **"it"** refers to the **trophy**, not the suitcase. A simple word-order rule might incorrectly assume that **"it"** refers to the nearest noun (**suitcase**), but self-attention uses the overall meaning and context to identify the correct reference.

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Lev
Technical / Concept	30%	Accurately explains tokenization, embeddings, and self-attention mechanics using correct terminology	Explains most mechanics correctly with minor gaps	Partial or superficial understanding of the mechanics	core
Application	25%	Correctly traces a worked example (e.g. computing attention relevance) step-by-step	Traces most steps correctly with minor errors	Significant errors when applying mechanics to the worked example	Can mec wor all
Quality	20%	Clear, logical, mathematically sound reasoning throughout	Mostly sound reasoning with minor errors	Reasoning has notable gaps or inconsistencies	Rea disc larg
Presentation	15%	Neat, well-labeled diagrams/working, easy to follow	Reasonably clear presentation of working	Presentation is unclear in places	Diff pres wo

Documentation / Communication	10%	Shows all work with clear justification for every step	Shows most work with adequate justification	Minimal work shown	No unju

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